(a)

Bulk Band (n_y, n_z) $(C_{2z}, \mathcal{I})$ $|\uparrow\rangle$ channel $|\downarrow\rangle$ channel $(C_{2z}, \mathcal{I})$

Energy

$\mathrm{CB}_2(1,1)$ $(-i, -1)$ $(+i, -1)$

$\mathrm{VB}_1(1,1)$ $(+i, +1)$ $(-i, +1)$

$\mathrm{VB}_1(2,1)$ $(-i, -1)$ $(+i, -1)$

$\mathrm{VB}_2(2,1)$ $(-i, -1)$ $(+i, -1)$

⟷ Electric-dipole coupling $e\mathcal{E}_y y$ ⟷ k_x-dependent coupling H_{k_x}

(b)

Energy

$\mathrm{VB}_1(1,1)$ $(+i, +1)$ $(-i, +1)$

$\mathrm{VB}_1(1,2)$ $(+i, -1)$ $(-i, -1)$

$\mathrm{VB}_1(2,1)$ $(-i, -1)$ $(+i, -1)$

$\mathrm{VB}_2(2,1)$ $(-i, -1)$ $(+i, -1)$

⟷ Electric-dipole coupling $e\mathcal{E}_z z$ ⟷ k_x-dependent coupling H_{k_x}